

Overview of Hierarchical clustering

Subject: Hierarchical clustering

1.

$$d(\mathbf{x}, \mathbf{y}) = 1 - \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

2.

Selecting the number of clusters In hierarchical clustering, the number of clusters is not determined automatically and must be chosen by analyzing the dendrogram. A common approach is to “cut” the dendrogram at a selected height, which partitions the data into clusters based on where the cut intersects the tree structure. The choice of cut level has a significant impact on the resulting clustering. One widely used heuristic is to identify large vertical gaps in the dendrogram, which correspond to substantial increases in the distance between successive merges. Cutting the dendrogram at a height just before such a gap can produce clusters that reflect natural groupings in the data. This is based on the observation that merges occurring at much larger distances often combine dissimilar clusters.

3.

Commercial implementations MATLAB includes hierarchical cluster analysis. SAS includes hierarchical cluster analysis in PROC CLUSTER. Mathematica includes a Hierarchical Clustering Package. NCSS includes hierarchical cluster analysis. SPSS includes hierarchical cluster analysis. Qlucore Omics Explorer includes hierarchical cluster analysis. Stata includes hierarchical cluster analysis. CrimeStat includes a nearest neighbor hierarchical cluster algorithm with a graphical output for a Geographic Information System.

4.

Formally, let (S, δ) be a metric space, and let A and B be nonempty compact subsets of S . The distance from a point a to a set B is defined as:

5.

Tradeoffs compared with traditional hierarchical clustering Compared with textbook agglomerative hierarchical clustering on a full $n \times n$ dissimilarity structure, HAL-x trades exact, purely geometric linkage updates for a pipeline that depends on embedding choices, density-thresholding heuristics, and the capacity of classifiers such as random forests or support vector machines to estimate separability in high dimensions. This design targets scalability: expensive steps are deliberately restricted to manageable downscaled subsamples, while label prediction can be applied broadly to the full dataset once the hierarchy and classifiers are established. Because of this, HAL-x is extremely useful when hierarchical clustering requires minimal RAM. A practical limitation, common to embedding-first pipelines, is that low-dimensional representations can distort global geometry and reduce accuracy. The HAL-x paper discusses when auxiliary preprocessing such as PCA may help for correlated features.

6.

The sum of all intra-cluster variance. The increase in variance for the cluster being merged (Ward's method) The probability that candidate clusters spawn from the same distribution function (V-linkage). In case of tied minimum distances, a pair is randomly chosen, thus being able to generate several structurally different dendrograms. Alternatively, all tied pairs may be joined at the same time, generating a unique dendrogram. One can always decide to stop clustering when there is a sufficiently small number of clusters (number criterion). Some linkages may also guarantee that agglomeration occurs at a greater distance between clusters than the previous agglomeration, and then one can stop clustering when the clusters are too far apart to be merged (distance criterion). However, this is not the case of, e.g., the centroid linkage where the so-called reversals (inversions, departures from ultrametricity) may occur.

7.

ALGLIB implements several hierarchical clustering algorithms (single-link, complete-link, Ward) in C++ and C# with $O(n^2)$ memory and $O(n^3)$ run time. ELKI includes multiple hierarchical clustering algorithms, various linkage strategies and also includes the efficient SLINK, CLINK and Anderberg algorithms, flexible cluster extraction from dendrograms and various other cluster analysis algorithms. Julia has an implementation inside the Clustering.jl package. Octave, the GNU analog to MATLAB implements hierarchical clustering in function "linkage". Orange, a data mining software suite, includes hierarchical clustering with interactive dendrogram visualisation. R has built-in functions and packages that provide functions for hierarchical clustering. SciPy implements hierarchical clustering in Python, including the efficient SLINK algorithm. scikit-learn also implements hierarchical clustering in Python. Weka includes hierarchical cluster analysis.

8.

Unlike Euclidean distance, it is scale-invariant and is useful when features are correlated or measured in different units. Hamming distance is used for categorical or binary data and counts the number of positions at which two observations differ. It is commonly applied to genetic sequences and binary feature vectors. The choice of metric interacts with the choice of linkage criterion. For example, Ward's method and centroid-based linkages are defined in terms of squared Euclidean distances, whereas single-linkage and complete-linkage clustering can be used with any valid metric.

9.

In this visualization, each point begins as its own cluster. At each step, the two clusters with the smallest distance between them are merged. On the left, dashed lines indicate which clusters are combined, and a running list of cluster groupings shows how clusters evolve over time. On the right, the dendrogram is constructed simultaneously, with each merge represented by a new branch. The height of each merge in the dendrogram corresponds to the distance or dissimilarity between the clusters being combined. Merges that occur at lower heights represent clusters that are more similar, while merges at higher heights indicate greater dissimilarity. Color-coding is used to associate each merge step in the data visualization with its corresponding branch in the dendrogram.

10.

$$d(\mathbf{x}, \mathbf{y}) = \frac{\|\mathbf{x} - \mathbf{y}\|}{\sqrt{S-1}} \quad \{\displaystyle d(\mathbf{x}, \mathbf{y}) = \frac{\|\mathbf{x} - \mathbf{y}\|}{\sqrt{S-1}}\}$$

11.

Interpreting dendrogram height A key aspect of interpreting a dendrogram is understanding the meaning of branch height. The height at which two clusters are joined represents the distance or dissimilarity between those clusters, where lower merges indicate greater similarity and higher merges indicate greater dissimilarity. Early merges at lower heights reflect the closest relationships between observations, while later merges combine increasingly different groups. When the dendrogram is scaled using actual or normalized distances, the vertical spacing between merges provides insight into the relative differences between clusters. Large vertical gaps between successive merges often indicate meaningful separations in the data and can help guide the selection of cluster boundaries, although dendrograms alone should not be used to definitively determine the number of clusters. Because dendrograms summarize underlying distance relationships, some information may be lost in the visualization, and interpretations are generally most reliable for clusters that merge at lower heights.